Please check that this question paper contains 09 questions and 02 printed pages within first ten minutes.

MORNING

27 FEB 2024

[Total No. of Questions: 09]

Uni. Roll No. ....

[Total No. of Pages: 02]

Program: B.Tech. (Batch 2018 onward)

Semester: 3rd

Name of Subject: Digital Electronics

Subject Code: ESCS-101

Paper ID: 16012

Scientific calculator is not allowed.

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

1) Parts A and B are compulsory
2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1. (a) Solve **$(614.15)_7$** into **$(X)_{10}$**

(b) List any two applications of multiplexer.

(c) Interpret the concept of Duality principle with an example.

(d) What is Race around condition? How it can be avoided?

(e) Find the value of X in **$(1101.101)_2 = (X)_{10}$**

(f) Compare performance of ECL and TTL.

Part – B

[Marks: 04 each]

Q2. Explain the working and truth table of master slave J-K flip flop in detail.

Q3. Compare and contrast Synchronous counter and Asynchronous counter?

Q4. Translate **C'D+ABC'+A'B'D+ABD'** into standard POS form.

Q5. Design a two bit magnitude comparator to compare equality and inequality.

Q6. Minimize the given expression using K-Maps. **$f = \pi M\,(1, 3, 6, 7, 8, 9, 12, 13, 17, 19, 22, 23, 24, 25, 30, 31)$**.

Q7. Explain block diagram of decoder to explain its functions.

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Part – C

[Marks: 12 each]

Q8. Construct a 4-bit Binary to Gray Code Converter with proper truth table, K-Maps and Circuit Diagram Implementation.

OR

Implement 4:1 MUX using 2:1 MUX.

Q9. Design a 3-bit Twisted Ring or Johnson counter using J-K flip flops to count the sequence of states with the help of truth table and timing diagram.

OR

Explain the following in detail:

(a) Programmable Logic Array (PLA),

(b) Programmable Array Logic (PAL),

(c) Field Programmable Gate Arrays (FPGA).

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